

Sealed Blind Predictions for Growth-Rate Observables: Energy-Flow Cosmology vs Λ CDM

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Symbiose Research

DOI: [10.6084/m9.figshare.32013156](https://doi.org/10.6084/m9.figshare.32013156)

License: CC-BY-4.0 Date: 2026-04-14

Abstract. We present pre-registered, falsifiable predictions from Energy-Flow Cosmology (EFC) for the growth-rate observable $f\sigma_8(z)$ and the Hubble distance ratio $D_H/r_d(z)$ at redshifts $z = 0.7$ and $z = 1.0$. Two independent MCMC chains (NUTS sampler) were frozen in February 2026 with cryptographic SHA-256 hashes and UTC timestamps, prior to the unblinding of the DESI DR2 full-shape redshift-space distortion (RSD) analysis. The key prediction— $f\sigma_8(z=0.7) = 0.430$ versus Λ CDM’s 0.449—represents a 2.0σ separation that will be directly testable once DESI publishes DR2 full-shape results. All baseline observational data (BOSS DR12, eBOSS DR16, DESI DR1) are included for reproducibility. This dataset constitutes a pre-registered cosmological experiment, not a confirmed result.

1. Introduction

The growth rate of large-scale structure, parameterized by $f\sigma_8(z)$, provides a direct probe of gravitational dynamics at cosmological scales. Unlike baryon acoustic oscillation (BAO) measurements, which test geometric distances, and Type Ia supernovae, which constrain the luminosity-distance relation, $f\sigma_8$ tests the *dynamical* growth of density perturbations—making it sensitive to modifications of General Relativity and non-standard dark-energy coupling.

Energy-Flow Cosmology (EFC) introduces an entropy-driven coupling parameter α that modifies the Hubble friction term in the growth equation. When fit to existing BAO and $f\sigma_8$ data via MCMC inference (NUTS sampler), EFC predicts systematically lower growth amplitudes at $z \lesssim 1$ compared to Λ CDM, along with distinct Hubble distance ratios at higher redshift.

These predictions were cryptographically sealed in February 2026—before the DESI DR2 full-shape RSD analysis is unblinded—to ensure they constitute genuine pre-registered tests rather than post-hoc curve fitting.

2. Sealed Predictions

Two MCMC freeze states were generated and hashed:

2.1 Primary Freeze: `freeze_20260218_050713`

Table 1: Primary sealed prediction parameters.

Parameter	Value
Sampler	NUTS
EFC coupling α	-0.689
SHA-256 hash (truncated)	<code>7a850cfa58477701...</code>
Sealed timestamp (UTC)	<code>2026-02-18T05:07:13</code>
$f\sigma_8$ crossover redshift	$z = 2.042$

Table 2: Predicted observables from primary freeze, with Λ CDM comparison.

Observable	Redshift	EFC	Λ CDM	Separation
$f\sigma_8$	$z = 0.7$	0.430	0.449	2.0σ
D_H/r_d	$z = 0.7$	19.797	20.719	2.3σ
D_H/r_d	$z = 1.0$	16.527	17.466	3.1σ

2.2 Secondary Freeze: `freeze_20260221_160857`

Table 3: Secondary sealed prediction parameters.

Parameter	Value
Sampler	NUTS
EFC coupling α	-0.702
SHA-256 hash (truncated)	<code>dbccda150abca0e7...</code>
Sealed timestamp (UTC)	<code>2026-02-21T16:08:57.818002</code>

Both freezes yield consistent $\alpha < 0$, indicating suppressed growth relative to Λ CDM across the tested redshift range.

3. Baseline Observational Data

The following $f\sigma_8$ measurements form the observational baseline used in the EFC inference pipeline. These are drawn from published BOSS, eBOSS, and DESI DR1 analyses.

Table 4: Compiled $f\sigma_8$ measurements used as input to EFC MCMC inference.

Survey	Tracer	z_{eff}	$f\sigma_8 \pm \sigma$	Reference
6dFGS	—	0.067	0.423 ± 0.055	Beutler+ 2012
BOSS DR12	LOWZ	0.320	0.427 ± 0.056	Alam+ 2017
BOSS DR12	CMASS	0.510	0.458 ± 0.038	Alam+ 2017
DESI DR1	LRGz0	0.510	0.449 ± 0.032	DESI 2024 V
BOSS DR12	CMASS	0.610	0.436 ± 0.034	Alam+ 2017
eBOSS DR16	LRG	0.700	0.473 ± 0.041	Gil-Marín+ 2020
DESI DR1	LRGz1	0.706	0.413 ± 0.029	DESI 2024 V
eBOSS DR16	ELG	0.850	0.315 ± 0.095	de Mattia+ 2021
DESI DR1	LRG+ELG	0.930	0.381 ± 0.034	DESI 2024 V
DESI DR1	ELG	1.317	0.396 ± 0.062	DESI 2024 V
eBOSS DR16	QSO	1.480	0.462 ± 0.045	Neveux+ 2020
DESI DR1	QSO	1.491	0.318 ± 0.076	DESI 2024 V

3.1 Preliminary Comparison at $z \approx 0.7$

The DESI DR1 LRGz1 measurement at $z_{\text{eff}} = 0.706$ provides an early (non-independent) check:

Table 5: Comparison of EFC and Λ CDM predictions against DESI DR1 at $z \approx 0.7$.

Model	Predicted $f\sigma_8$	$ \Delta $ from DR1
EFC ($\alpha = -0.689$)	0.430	+0.017
Λ CDM	0.449	+0.036
DESI DR1 LRGz1	0.413 ± 0.029	—

EFC is closer to the measured value, but the difference is $< 1\sigma$ and does not constitute a decisive test. The definitive test requires DESI DR2 full-shape RSD data with improved statistical precision and independent systematics.

4. Discussion

4.1 Interpretation

The sealed predictions test a specific physical mechanism: EFC’s entropy-driven modification of the Hubble friction term, parameterized by coupling α . A negative α suppresses structure growth at late times ($z \lesssim 1$), producing systematically lower $f\sigma_8$ values than Λ CDM.

All three observables (Table 2) point in the same direction—lower growth amplitude, lower D_H/r_d —which reduces the probability that the signal is due to noise or parameter degeneracy. This multi-observable consistency is the strongest aspect of the prediction set.

4.2 Limitations

Several caveats apply:

- The $2.0\text{--}3.1\sigma$ separations are suggestive, not decisive. A 5σ detection-level signal would require either a single high-precision measurement or joint constraints across multiple redshift bins.
- RSD measurements are sensitive to bias modeling, non-linear corrections, and tracer selection effects. DR2 systematics could shift the central values.
- No independent pipeline replication has been performed. The inference relies on the EFC MCMC implementation within the Symbiose framework.
- This is a pre-registered test, not a confirmed detection.

4.3 What DR2 Must Deliver

For this prediction to be *confirmed*:

- DESI DR2 full-shape $f\sigma_8(z \approx 0.7)$ must fall within $\sim 1\sigma$ of 0.430.
- Consistency should hold across multiple tracers (LRG, ELG).
- The D_H/r_d predictions at $z = 0.7$ and $z = 1.0$ must independently trend below Λ CDM expectations.

For *falsification*: a DR2 value consistent with Λ CDM’s 0.449 at $< 1\sigma$ would require revision of the EFC growth-sector parameterization.

5. Reproducibility

5.1 Data Availability

All data required to reproduce the baseline comparison are included in the accompanying JSON files:

- `sealed_predictions.json` — SHA-256 hashes, timestamps, and all predicted values.
- `rsd_measurements.json` — complete $f\sigma_8$ compilation with survey metadata and references.

5.2 Cryptographic Verification

The SHA-256 hashes in Table 1 and Table 3 were computed over the full MCMC chain state (parameter vectors, log-posteriors, and sampler diagnostics) at freeze time. The truncated hashes shown here allow verification against the full hashes stored in the Symbiose knowledge graph.

5.3 Code and Infrastructure

The EFC MCMC inference pipeline runs within the Symbiose autonomous research framework. Source code and configuration are available at:

<https://github.com/upertedai/EFC>

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